

Assignment 6: Implement Bully and Ring algorithm for leader election.

Bully Algorithm Steps (Java)

Step 1: Open VS Code

Install Java Extension Pack.

Step 2: Create Java File

Create file:

Bully.java

Paste the Bully Algorithm Java code and save the file.

Step 3: Compile Program

Open terminal in VS Code and run:

```
javac Bully.java
```

Step 4: Run Program

```
java Bully
```

Ring Algorithm Steps (Java)

Step 1: Create Java File

Create file:

Ring.java

Paste the Ring Algorithm Java code and save the file.

Step 2: Compile Program

```
javac Ring.java
```

Step 3: Run Program

```
java Ring
```

Viva Questions and Answers for Bully and Ring Algorithm

1. What is Leader Election Algorithm?

Leader election algorithm selects one process as coordinator/leader in a distributed system.

2. What is a coordinator process?

Coordinator is the process that controls and manages distributed activities.

3. Why is leader election needed?

Leader election is needed when the current coordinator fails or crashes.

4. What is Bully Algorithm?

Bully Algorithm elects the process with the highest process ID as coordinator.

5. Why is it called Bully Algorithm?

Because higher priority processes dominate lower priority processes.

6. What is Ring Algorithm?

Ring Algorithm arranges processes in a logical ring and passes election messages around the ring.

7. Which process becomes leader in both algorithms?

The process with the highest ID becomes leader.

8. What happens if coordinator crashes?

A new election process starts automatically.

9. What is process ID?

A unique number assigned to each process.

Example:

1, 2, 3, 4, 5

10. How does Bully Algorithm work?

- Process detects coordinator failure
 - Sends election message to higher ID processes
 - Highest active process becomes coordinator
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11. How does Ring Algorithm work?

- Election message circulates in ring structure
 - Highest process ID is selected as leader
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12. Which algorithm is faster?

Bully Algorithm

13. Which algorithm has less message overhead?

Ring Algorithm

14. What is message overhead?

Number of messages exchanged during communication.

15. What is distributed system?

A distributed system is a group of computers/processes working together.

16. Advantages of Bully Algorithm

- Fast election
- Easy implementation

17. Disadvantages of Bully Algorithm

- High message overhead
- Not efficient for large systems

18. Advantages of Ring Algorithm

- Organized communication
- Less message collision

19. Disadvantages of Ring Algorithm

- Slower than Bully Algorithm
- Ring failure affects communication

20. What is ring topology?

A network structure where processes are connected in circular form.

21. What is the output of your Bully Algorithm program?

Coordinator Process is: 5

New Coordinator is: 4

22. What is the output of your Ring Algorithm program?

Leader elected is: 4

23. Which language is used in your program?

Java

24. Real-life applications of leader election

- Distributed databases
- Cloud computing
- Distributed operating systems
- Network servers

25. Difference between Bully and Ring Algorithm

Bully Algorithm	Ring Algorithm
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Faster	Slower
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More messages	Fewer messages
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Direct election	Ring communication
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